ETFs Simulation with Temporal Memory: A Positive Fractional–Diffusive Volatility Model

Enia Ardid, Pablo Casado de Amezúa

Research manuscript — 22 October 2026

Abstract

We study a positive conditional-variance model combining finite fractional-integral memory and implicit diffusion on a correlation graph. We establish positivity, fixed-parameter moment stability and exact multi-step variance forecasts for the implemented recursion, without deriving it from an option-pricing equation. A chronological study of the S&P 500 and eleven sector ETFs uses 186 origins during 2023–2026 and horizons of 1, 5 and 21 observations. Adding persistence, asymmetry and multi-horizon validation improves the initial fractional model in all six panel–horizon comparisons, but GJR-GARCH remains stronger. These development results are exploratory. A further protocol fixes eight country ETFs before download and retains the model unchanged. It adds FIGARCH and a fixed 50:50 fractional/GJR forecast ensemble. The ensemble improves pooled QLIKE over the standalone fractional model and FIGARCH in all nine comparisons across the three panels. It exceeds GJR-GARCH in only three, with all paired ranges against GJR including zero. The country panel does not confirm a general diffusion benefit. A subsequent Network GARCH comparison favors the fractional model in the country panel and the weighted network benchmark in the sector panel, with paired ranges including zero. The contribution is an explicit constrained fractional–diffusive risk model and a reproducible assessment of its properties, forecast combinations and limitations on retrospective adjusted-price data.

Keywords: fractional memory; conditional variance; graph diffusion; sector ETFs; volatility forecasting; chronological validation.

Data and code availability. Code, tests, protocols and aggregate results accompany this draft. The project repository is https://github.com/enia-ardid/mathematical_finance. The accompanying code and manuscript should be identified by their Git commit. Raw prices are retained locally with source metadata and hashes and are not bundled.

Research manuscript for author review; not a peer-reviewed publication.

Contents

1 Introduction **2**
1.1 Position relative to network volatility models . . . 2

2 Data, notation and prediction target **3**
2.1 Observed prices and returns . . . 3
2.2 Risk target and common mean convention . . . 3

3 A positive fractional–diffusive model **4**
3.1 Finite fractional-integral memory . . . 4
3.2 Correlation graph and implicit diffusion . . . 4
3.3 Conditional variance and return generation . . . 5

4 Mathematical properties **6**

5 Forecasting and estimation **7**
5.1 Exact multi-step marginal moments . . . 7
5.2 Chronological selection . . . 7
5.3 Benchmarks and ablations . . . 8

6 Evaluation of the initial specification **8**
6.1 Loss and dependence . . . 8
6.2 Pooled comparison . . . 9
6.3 Variation over time . . . 9

7 Persistence and asymmetric innovations **11**
7.1 Specification and moment properties . . . 11
7.2 Effective memory and graph modes . . . 12
7.3 Recorded follow-up protocol . . . 12
7.4 Follow-up results . . . 13

8 Extension to country ETFs and additional benchmarks **14**
8.1 A specification fixed before the new download . . . 14
8.2 Additional forecaster and fractional benchmark . . . 15
8.3 Results . . . 15
8.4 Direct comparison with Network GARCH . . . 16
8.5 Observed evolution and rolling predictions . . . 18

9 Interpretation, limitations and conclusions **21**

A Implementation and reproduction **21**

B Annual scores **22**

C Relation to the original model **23**

D Observed and predicted paths for every ETF **24**

1 Introduction

Financial returns provide a difficult setting for directional forecasting: recent prices already constitute a strong reference, while a visually plausible price path can conceal weak predictive information. A more focused question is whether temporal memory and sector interactions help forecast the magnitude of future fluctuations. This paper retains those two ingredients and assigns them to an explicit conditional-variance model.

Fractional filters offer a flexible way to weight past observations. Fractional differencing [1] and FIGARCH [2] provide established time-series constructions with slowly decaying dependence. Here we use a finite, normalized fractional-integral kernel to summarize past squared innovations. Interactions between assets are represented by a correlation graph, so the diffusion operator acts on an estimated network rather than an ordering of sector names. Graph signal processing provides the corresponding operator framework [3].

Spatial ARCH models already allow conditional heteroskedasticity to depend on neighboring locations [4]. For financial networks, network log-ARCH models combine temporal effects with contemporaneous network interactions [5]. Our specification instead applies a positive graph resolvent to lagged squared innovations with a shared finite fractional kernel. The resulting model is a restricted multivariate ARCH specification: its lag matrices share a spatial operator and a common temporal profile.

Our model applies a normalized finite fractional integral to squared standardized innovations, then smooths this signal through a correlation-graph resolvent. An explicit positive intercept and a feedback constraint control the resulting variance recursion. The model is specified directly in discrete time and targets conditional variance. This gives the memory and diffusion terms a precise role while retaining a small set of interpretable parameters.

We develop this specification, establish its positivity and fixed-parameter moment properties, and derive multi-step conditional variances without simulation error. The empirical study then asks whether the fractional profile and graph diffusion improve forecasts at daily, weekly and monthly horizons. Comparisons include conventional volatility models, variants with individual components removed, and a separately recorded Network GARCH benchmark. Code, protocols and aggregate results accompany the study.

1.1 Position relative to network volatility models

Network GARCH already uses graph structure to reduce the dimension of multivariate volatility estimation, with stationarity and QMLE results [6]. Threshold network GARCH also incorporates asymmetric shock responses [7]. More recently, generalized network GARCH includes neighboring shock and variance terms alongside covariance forecasts [8]. Neither network dependence nor asymmetry is original to the present specification. Relative to these models, our restriction is a common positive resolvent at every lag and a normalized fractional temporal profile. Relative to FIGARCH, our finite normalized kernel and subunit feedback define a different recursion. Section 8 includes a fitted FIGARCH reference, and Section 8.4 adds the first-order Network GARCH recursion of Zhou et al. [6], adapted to the available correlation networks. Network log-ARCH, threshold Network GARCH and DCC-GNGARCH remain outside the fitted comparison. Our mathematical

arguments specialize positive-filter results to the shared temporal and spatial structure; the effective impulse response below makes that restriction explicit.

The initial specification improves average losses relative to historical variance and EWMA, but GARCH remains stronger. A persistent asymmetric refinement improves all six pooled comparisons with that first specification and exceeds GARCH at one and five observations in both US panels. GJR-GARCH remains stronger throughout that standalone comparison. The additional external-universe study and fixed forecast combination extend the assessment in Section 8. Diffusion helps the one-session sector average modestly, with uncertainty ranges including zero. These comparisons distinguish gains from persistence and asymmetry from the less consistent contribution of the memory profile and graph.

2 Data, notation and prediction target

2.1 Observed prices and returns

Let $P_{t,j} > 0$ denote the closing price of asset j at retained trading observation t, and define

$$r_{t,j} = \log(P_{t,j}/P_{t-1,j}). \tag{1}$$

The information available at forecast origin o is $\mathcal{F}_o = \sigma(P_0, \ldots, P_o)$. All transformations, graph estimates and parameter choices used to forecast from o depend only on this information.

The market snapshot was downloaded on 17 October 2026 from Yahoo Finance through `yfinance` 0.2.66, with adjusted prices and an exclusive request end date of 17 October 2026. It contains the index `^GSPC` and the eleven ETFs XLB, XLE, XLF, XLI, XLK, XLP, XLRE, XLU, XLV, XLY and XLC. After inception trimming and alignment, common prices run from 19 June 2018 through 16 October 2026. Prices must be positive, finite, uniquely dated and increasing in date; missing interior observations are not filled.

The index and ETF panel are fitted separately. Adjusted ETF returns and an index price-return series are not identical economic objects; we do not interpret their combination as a self-financing index replication. The snapshot is retrospective and may contain adjustment revisions unavailable at historical forecast dates. Chronological code prevents access to future rows, but it does not turn revised data into point-in-time observations.

2.2 Risk target and common mean convention

At each parameter refit, estimate a mean μ_j and a positive scale s_j^2 from the observed fitting window. These estimates are shared by all benchmark models until the next refit. Write

$$u_{t,j} = r_{t,j} - \mu_j, \qquad z_{t,j} = u_{t,j}^2/s_j^2. \tag{2}$$

The h-observation evaluation target is

$$Y_{o,h,j} = \sum_{\ell=1}^{h} (r_{o+\ell,j} - \mu_j)^2, \qquad h \in \{1, 5, 21\}. \tag{3}$$

This is a daily squared-return proxy, not intraday realized variance. Under the specified constant conditional mean it is conditionally unbiased for the sum of future conditional variances. If the fitted mean differs from the true conditional mean, the target also contains the corresponding squared mean error. We therefore interpret the empirical task conservatively as forecasting centered return second moments.

3 A positive fractional–diffusive model

3.1 Finite fractional-integral memory

For $0 < d \leq 1$, the binomial expansion of $(1 - B)^{-d}$ yields positive coefficients

$$g_0(d) = 1, \qquad g_k(d) = \frac{k - 1 + d}{k}\, g_{k-1}(d) = \frac{\Gamma(k + d)}{\Gamma(d)\Gamma(k + 1)}. \tag{4}$$

With memory length M, normalize these coefficients:

$$p_k(d) = \frac{g_k(d)}{\sum_{v=0}^{M-1} g_v(d)}, \quad m_t = \sum_{k=0}^{M-1} p_k(d) z_{t-1-k}. \tag{5}$$

Thus $p_k > 0$ and $\sum_k p_k = 1$. For large k, $g_k(d)$ behaves as $k^{d-1}/\Gamma(d)$. Within a fixed window, increasing d spreads weight toward older observations; $d = 1$ gives a uniform moving average. The study fixes $M = 252$. It does not infer d from a Hurst estimator.

The normalized operator is a finite-window analogue of a fractional integral. For a smooth signal q, a physical window T and a refining observation step, the corresponding normalized fractional integral is

$$\mathcal{I}_{d,T} q(t) = \frac{d}{T^d} \int_0^T a^{d-1} q(t - a)\, da. \tag{6}$$

This relation motivates the kernel; the empirical model itself is the finite discrete recursion (5). Finite memory and subunit feedback do not establish asymptotic long-range dependence or convergence to a fractional stochastic PDE.

3.2 Correlation graph and implicit diffusion

Estimate a Ledoit–Wolf covariance matrix [9] after standardizing the fitting-window returns. Convert it to a correlation matrix R and set

$$W_{ij} = \max(R_{ij}, 0)^2 \quad (i \neq j), \qquad W_{ii} = 0. \tag{7}$$

Let $D = \mathrm{diag}(W\mathbf{1})$ and rescale the graph Laplacian by its mean degree:

$$L = \frac{D - W}{\max(J^{-1} \operatorname{tr} D, 10^{-12})}, \qquad A_\tau = (I + \tau L)^{-1}, \quad \tau \geq 0. \tag{8}$$

For the scalar index, $L = 0$ and $A_\tau = 1$; index results contain no evidence about cross-sector diffusion. For sectors, the update $x = A_\tau m$ is the implicit Euler step $x - m = -\tau L x$

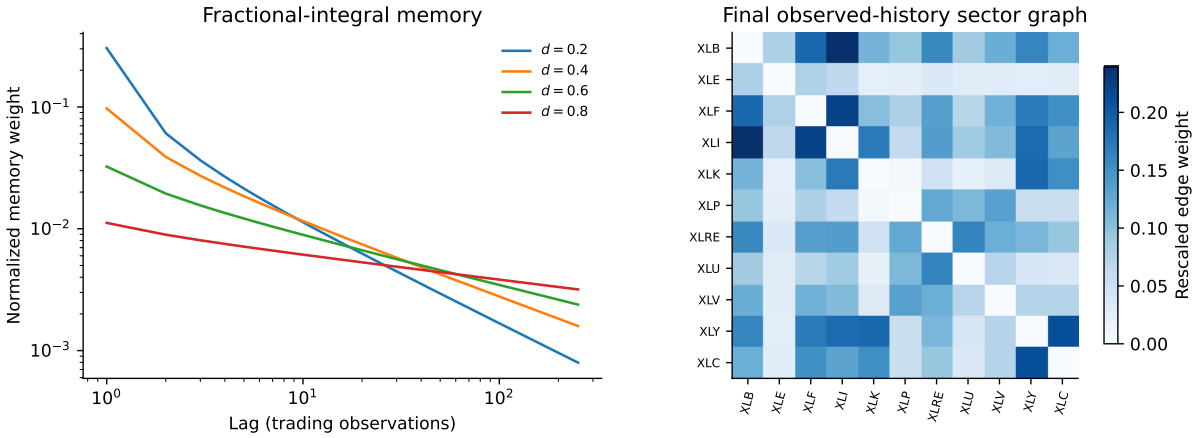


Figure 1: Normalized fractional-integral weights and the sector graph from the last parameter refit in the recorded backtest. The graph uses only the history available at that refit. Weights are shown over the actual 252-observation memory window; an infinite-memory interpretation is not assumed.

for graph heat diffusion. The diffusion parameter controls smoothing on this estimated network, not a causal contagion rate.

No arbitrary one-dimensional sector ordering is required. Under a permutation matrix Π, the same construction transforms L into $\Pi L \Pi^{\mathsf{T}}$ and A_τ into $\Pi A_\tau \Pi^{\mathsf{T}}$. Forecasts are consequently permutation equivariant.

3.3 Conditional variance and return generation

For $0 \le \eta < 1$, define normalized conditional variance h_t and asset-scale variance v_t by

$$h_t = (1-\eta)\mathbf{1} + \eta A_\tau m_t, \qquad v_t = s^2 \odot h_t, \tag{9}$$

where $\odot$ denotes elementwise multiplication. Returns obey

$$r_t = \mu + \operatorname{diag}(\sqrt{v_t})\,\varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0, \quad \mathbb{E}[\varepsilon_{t,j}^2 \mid \mathcal{F}_{t-1}] = 1. \tag{10}$$

For simulations, the implementation uses temporally independent Gaussian innovation vectors with correlation R. Cross-sectional correlation does not invalidate the marginal moment results below. Gaussian innovations are a simulation assumption, not a claim that market return distributions are Gaussian.

Diffusion acts on standardized variance signals. It preserves their sum, not necessarily the sum of variances in original asset units. This distinction avoids forcing low-volatility and high-volatility sectors to have equal scale. The positive intercept is explicit, and all simulated paths use the same stated recursion rather than a fitted terminal price correction.

To make the connection with multivariate ARCH explicit, let $S = \operatorname{diag}(s_1^2, \ldots, s_J^2)$. In asset units, the lag coefficient matrices are $C_k = \eta p_k S A_\tau S^{-1}$, so that $v_t = (1-\eta)s^2 + \sum_{k=0}^{M-1} C_k(u_{t-1-k} \odot u_{t-1-k})$. All C_k are nonnegative, and $\rho(\sum_k C_k) = \eta$. Fractional weighting and graph diffusion constrain these matrices to an interpretable shared structure.

4 Mathematical properties

Proposition 1 (Positive, nonamplifying graph diffusion)**.** *For a symmetric nonnegative adjacency matrix and $\tau \geq 0$, A_τ is entrywise nonnegative and doubly stochastic. Its eigenvalues lie in $(0, 1]$. For any real vector x,*

$$\min_j x_j \leq (A_\tau x)_i \leq \max_j x_j, \qquad (A_\tau x)^\mathsf{T} L (A_\tau x) \leq x^\mathsf{T} L x. \tag{11}$$

Proof. The Laplacian is symmetric positive semidefinite and $L\mathbf{1} = 0$. Hence $I + \tau L$ is invertible and $A_\tau \mathbf{1} = \mathbf{1}$. For $b \geq 0$, a negative minimum of the solution to $x + \tau L x = b$ would make the corresponding left-hand component negative, a contradiction. Thus $A_\tau \geq 0$. Symmetry gives unit column sums. The componentwise bound follows from convex averaging. Diagonalizing L gives eigenvalues $(1 + \tau\lambda_i)^{-1}$ for A_τ; the energy inequality follows from $\lambda_i/(1 + \tau\lambda_i)^2 \leq \lambda_i$. □

Proposition 2 (Positivity and fixed-parameter moment stability)**.** *Fix $M < \infty$, $s_j^2 > 0$ and $0 \leq \eta < 1$. For nonnegative observed squared innovations, $v_{t,j} \geq (1 - \eta)s_j^2$. If the innovation vectors are iid over time with zero means and unit marginal second moments, the normalized squared-return recursion admits a unique causal stationary solution with finite first moments, and its stationary marginal expectations satisfy $\mathbb{E}z_t = \mathbf{1}$. Consequently, the corresponding stationary returns have variance s_j^2 and finite second moments $\mu_j^2 + s_j^2$.*

Proof. The lower bound follows from nonnegative weights and nonnegative diffusion. If initial component expectations are bounded by $C \geq 1$, then $\mathbb{E}h_{t,j} \leq (1 - \eta) + \eta C \leq C$, and $\mathbb{E}z_{t,j} = \mathbb{E}h_{t,j}$. For two histories driven by the same innovation sequence, put $e_t = \mathbb{E}\|z_t - z'_t\|_1$. Independence of the current innovations from the past, unit marginal second moments and the unit column sums of A_τ give

$$e_t \leq \eta \sum_{k=0}^{M-1} p_k e_{t-1-k}. \tag{12}$$

The largest expected discrepancy contracts by at least η each time an entire block of M initial lags is replaced. Backward iteration from a bounded deterministic history is therefore Cauchy in L^1, yielding a causal stationary limit; the same bound gives uniqueness among finite-first-moment solutions. In stationarity, $b = \mathbb{E}z_t$ solves $b = (1 - \eta)\mathbf{1} + \eta A_\tau b$. Since $\rho(\eta A_\tau) < 1$, its unique solution is $b = \mathbf{1}$. □

These results concern fixed parameters. Rolling estimates need not form a stationary process, and finite second moments do not imply finite fourth moments, asymptotic normality of loss averages, or guaranteed forecast accuracy. The finite-window contraction also explains why a power-shaped kernel should not be advertised as a proof of infinite-horizon long memory.

5 Forecasting and estimation

5.1 Exact multi-step marginal moments

At origin o, let $\bar{z}_{o+\ell|o}$ equal observed z for nonfuture lags, and recursively replace future squared innovations by their conditional expectations:

$$\bar{h}_{o+\ell|o} = (1-\eta)\mathbf{1} + \eta A_\tau \sum_{k=0}^{M-1} p_k \bar{z}_{o+\ell-1-k|o}, \qquad \bar{z}_{o+\ell|o} = \bar{h}_{o+\ell|o}. \tag{13}$$

Linearity and unit innovation second moments make this recursion exact under the fitted variance model. The aggregate forecast is

$$\widehat{V}_{o,h} = s^2 \odot \sum_{\ell=1}^{h} \bar{h}_{o+\ell|o}. \tag{14}$$

With the model's conditional mean, this also gives the marginal conditional variance of cumulative future log-returns because the centered increments are martingale differences. It does not provide an exact Gaussian predictive law for their sum: future variances are themselves random. The study evaluates variance forecasts directly and does not exponentiate an assumed Gaussian multi-step law into a purported exact expected price.

5.2 Chronological selection

Forecast origins occur every five retained observations on a grid beginning at price position 252; scoring begins in 2023. At the first scored origin and whenever at least 21 observations have elapsed since the last refit, parameters are updated. This gives a 25-observation refit spacing on the five-observation grid, with 38 refits per panel in the reported experiment.

At a refit origin, reserve the final 252 observed returns for validation. Means, scales and graph structure used to compute validation predictions are estimated strictly before that block, using at most 1,008 returns. As validation advances, already observed squared returns may update the memory state. Their future counterparts are never accessed. Select the minimum pooled one-step QLIKE over

$$d \in \{0.2, 0.4, 0.6, 0.8\}, \qquad \eta \in \{0.8, 0.95, 0.99\}, \tag{15}$$

$$\tau \in \{0, 0.25, 1, 4\}. \tag{16}$$

The index fixes $\tau = 0$. Stable candidate order resolves exact ties. After selection, refit means, scales and the graph on the last 1,008 available returns, including validation observations, and forecast subsequent targets. This is a chronological selection design, not an untouched final test: the historical period was available during earlier research iterations.

The same selected parameters serve all three horizons. Consequently, favorable one-step validation does not guarantee optimal 21-step predictions. All methods are scored at the same origins, excluding end blocks without complete targets for every horizon.

5.3 Benchmarks and ablations

1. **Historical variance:** the current fitting-window sample variance, held constant over each forecast horizon.
2. **EWMA:** squared-innovation averaging with decay selected from $\{0.9, 0.94, 0.97, 0.99\}$ on the same validation block. Its future conditional variance remains constant under its own update equation.
3. **GARCH(1,1):** independent asset models estimated by Gaussian quasi-maximum likelihood [10]. Positivity is imposed and $\alpha+\beta < 0.999$. Two fixed initializations are used; all 456 selected asset-level fits report convergence in this run. No benchmark fit is replaced silently by a weaker model.
4. **Fractional only:** the proposed kernel and feedback with $\tau = 0$, with its own validation-selected d and η.
5. **Exponential plus graph:** replace the fractional average by EWMA while retaining positive feedback and graph diffusion. Its decay, feedback and diffusion grids have the same candidate count as the fractional graph model.

The last two comparisons distinguish the contribution of graph smoothing from the shape of the memory kernel. They are stronger checks than comparing only against a constant price or historical variance.

6 Evaluation of the initial specification

6.1 Loss and dependence

The primary score is a normalized QLIKE loss,

$$\mathcal{L}_{o,h,j} = \log\left(\frac{\widehat{V}_{o,h,j}}{hs_j^2}\right) + \frac{Y_{o,h,j}}{\widehat{V}_{o,h,j}}. \tag{17}$$

The normalization is known at the origin and common to all methods. It changes the level, not paired comparisons. QLIKE is appropriate for conditional second-moment forecasting with noisy, conditionally unbiased proxies [11]; that robustness does not remove sampling noise or mean-model error. Scores can be negative, and ratios of score levels are not reported as percentage improvements. A normalized squared-error diagnostic is also retained in the machine-readable results.

There are 186 common origins from 9 January 2023 to 17 September 2026, with targets through 16 October 2026. The sector panel contributes 2,046 asset–origin pairs per horizon, which must not be treated as independent replications. Twenty-one-observation targets overlap. Paired circular block-bootstrap ranges use 2,000 replicates, seed 1729, and blocks of five or ten origins. Entire sector vectors are resampled together. These are nominal descriptive ranges; they do not correct for all model selection, multiple comparisons, nonstationarity or previous researcher exposure.

panel	index			sectors		
horizon	1	5	21	1	5	21
Historical	0.8368	0.6695	0.6737	0.8167	0.7187	0.7222
EWMA	0.8397	0.4576	0.5602	0.7195	0.5914	0.6251
GARCH	0.6004	0.3640	0.4714	0.6297	0.5499	0.6021
Fractional only	0.6514	0.4223	0.5467	0.6549	0.5638	0.6175
Exponential + graph	0.6690	0.3767	0.4841	0.6524	0.5542	0.5953
Fractional + graph	0.6514	0.4223	0.5467	0.6405	0.5627	0.6196

Table 1: Mean normalized QLIKE at 1, 5 and 21 observations. Lower is better. All methods use identical origins and the same mean convention. The index has no cross-sector diffusion, so its two fractional rows coincide.

6.2 Pooled comparison

Table 1 shows lower mean loss for fractional diffusion than tuned EWMA at every panel–horizon combination. The one-session differences are -0.18834 for the index and -0.07899 for sectors. Nevertheless, GARCH has lower mean QLIKE in all six cases. For sectors, the proposed method's excess losses relative to GARCH are 0.01084, 0.01287 and 0.01752 across the three horizons. These findings support a useful alternative representation, not dominance over standard conditional-variance models.

Graph diffusion changes the one-session sector score from 0.65489 to 0.64053. Its gain is much smaller at five observations, and the no-diffusion model is slightly better at 21 observations. Nonzero diffusion is selected at 31 of 38 sector refits, but validation selection is not itself evidence of outer-sample predictive benefit. The exponential graph model outperforms fractional diffusion at five and 21 observations in both panels; the fractional kernel has the lower one-session mean in both panels. Memory shape matters, but no shape dominates all horizons.

With ten-origin blocks, the nominal 95% range for the one-session sector difference from EWMA is $[-0.26077, 0.03470]$. Relative to no diffusion, it is $[-0.03672, 0.00145]$. Both include zero. The difference from GARCH is 0.01084, with range $[-0.03822, 0.04668]$. We therefore do not infer a reliable edge over these dynamic references. Full ranges for both block lengths are distributed with the generated tables.

6.3 Variation over time

Average improvements are not uniform over calendar years. For the index, the proposed method loses to EWMA at all horizons in 2023, wins at all horizons in 2024 and 2026, and has mixed results in 2025. For sectors, it loses at all horizons in 2023 and improves on EWMA in the later years. The 2025 index one-session difference of -0.5367 materially influences the pooled average; it should not be interpreted as a stable gain in every regime.

Figure 3 illustrates the response to changing fluctuation levels. The plot is descriptive and supplements the numerical losses; visual agreement alone is not a forecast test. Annual scores and losses for every asset are recoverable from the origin-level local outputs, while aggregate annual tables are included in the manuscript bundle.

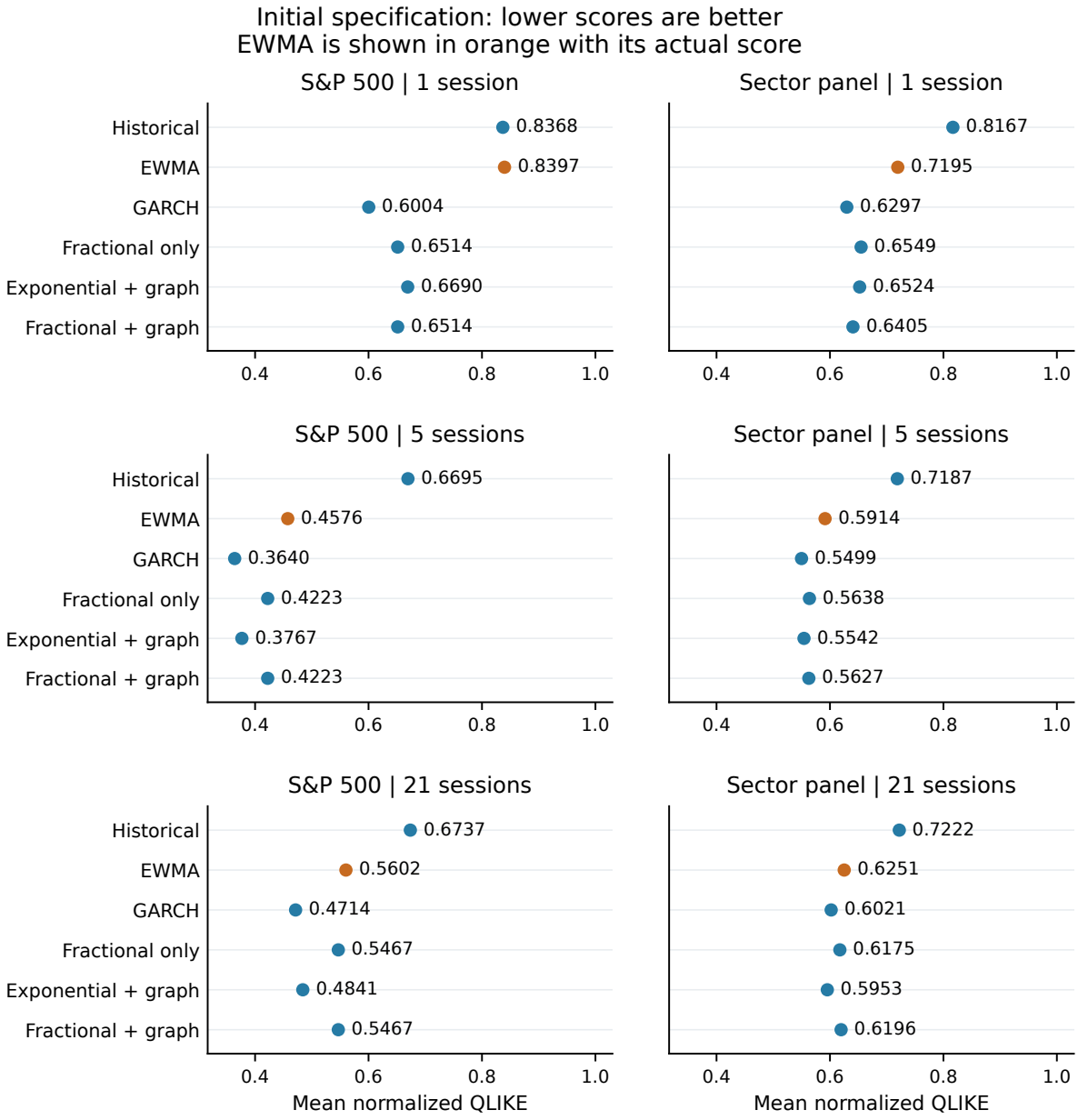


Figure 2: Mean normalized QLIKE for the initial specification and references. Lower values indicate smaller loss. Each point and its numerical label show the actual score, including EWMA in orange. The six panels share the same horizontal scale and reproduce Table 1; all methods use identical origins within each panel and horizon.

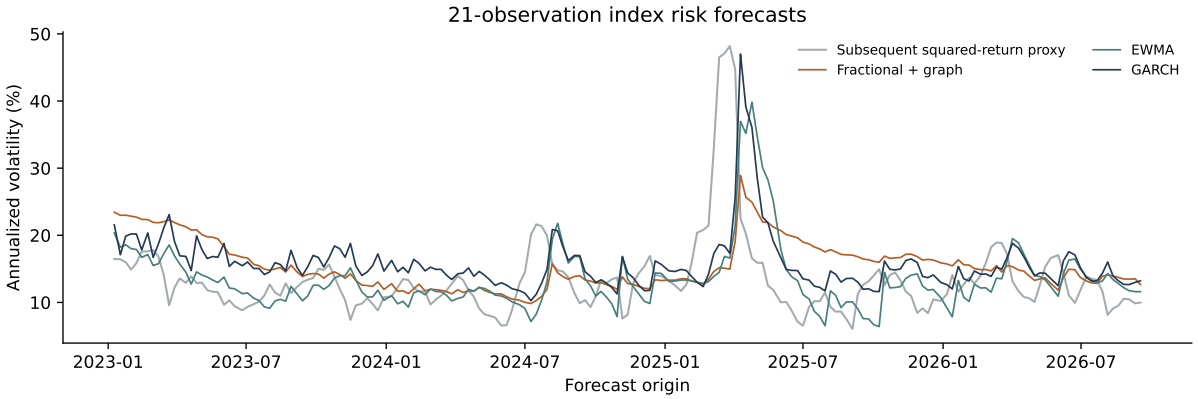


Figure 3: Index risk forecasts over 21 subsequent observations, displayed in annualized volatility units. The gray series is the subsequently observed squared-return proxy, plotted at its forecast origin for comparison; it was not available then. The proxy is noisy and is not the latent conditional variance.

7 Persistence and asymmetric innovations

7.1 Specification and moment properties

The initial model responds identically to positive and negative innovations and selects parameters using only one-step losses. We retain its fractional kernel and graph diffusion while allowing a persistent variance state and asymmetric shock contributions. Define

$$x_{t,j} = z_{t,j}\frac{1+\delta\mathbf{1}\{u_{t,j}<0\}}{1+\delta/2}, \tag{18}$$

$$h_t = (1-b)(1-\eta)\mathbf{1} + bh_{t-1} + (1-b)\eta A_\tau \sum_{k=0}^{M-1} p_k x_{t-1-k}, \tag{19}$$

with $v_t = s^2 \odot h_t$, $0 \le b < 1$ and $\delta \ge 0$. The parameter b smooths the variance response. With $\delta = 1$, a negative innovation has twice the contribution of a positive innovation of the same magnitude.

For the asymmetric recursion, assume conditionally symmetric innovations with unit second moments. This is stronger than a zero-mean, unit-variance assumption: symmetry gives $\mathbb{E}[\varepsilon_{t,j}^2\mathbf{1}\{\varepsilon_{t,j}<0\}] = 1/2$ and therefore $\mathbb{E}[x_t \mid \mathcal{F}_{t-1}] = h_t$. Gaussian simulation innovations satisfy the assumption; it is not asserted as an empirical property of market shocks. Future expected variances are obtained by the same substitution as in Section 5.1, retaining the additional state h_{t-1}.

Unit initial variance and unit presample transformed squares give the floor $v_{t,j} \ge (1-\eta)s_j^2$ by induction. For fixed parameters and iid symmetric innovations, the shared-noise argument for Proposition 2 now gives, for discrepancies between normalized variance processes,

$$e_t \le be_{t-1} + (1-b)\eta \sum_{k=0}^{M-1} p_k e_{t-1-k}, \qquad b + (1-b)\eta < 1. \tag{20}$$

The maximum expected discrepancy over the finite lag state contracts over successive M-observation blocks. The same backward-iteration argument yields a unique causal finite-

first-moment stationary variance process. Its mean is $\mathbf{1}$, so return variance is s_j^2. The recursive state adds exponentially decaying persistence; it does not establish asymptotic long-range dependence.

7.2 Effective memory and graph modes

The persistence state has an exact filter interpretation. For fixed parameters, unroll (19) after removing its initialization transient to obtain

$$h_t = (1-\eta)\mathbf{1} + \eta A_\tau \sum_{j=1}^{\infty} \widetilde{p}_j x_{t-j}, \tag{21}$$

$$\widetilde{p}_j = (1-b) \sum_{k=0}^{\min(M-1,j-1)} p_k b^{j-1-k}. \tag{22}$$

Indeed, repeated substitution expands the state into a geometric sum; grouping terms with the same innovation lag yields (22). Since all terms are nonnegative, summing in either order gives $\sum_{j\geq 1} \widetilde{p}_j = (1-b)\sum_k p_k \sum_{\ell\geq 0} b^\ell = 1$. For $0 < b < 1$ and $j \geq M$, $\widetilde{p}_{j+1} = b\widetilde{p}_j$. At $b = 0$ the weights reduce to p_{j-1} for $1 \leq j \leq M$, with zero tail. Thus the refinement geometrically smooths the finite fractional kernel; its infinite tail, when present, is exponential rather than hyperbolic.

With $L = U\Lambda U^\mathsf{T}$, the graph-mode coefficient at lag j is $\eta\widetilde{p}_j/(1+\tau\lambda_\ell)$. Temporal weighting and spatial smoothing are separable: every lag shares the same graph attenuation. This clarifies both interpretability and a restriction relative to more general network lag matrices. A numerical impulse-response test verifies the identity against the implemented filtering recursion.

7.3 Recorded follow-up protocol

This refinement was designed after inspecting the initial study. The separate `paper/refinement_protocol.json` records its rules before its candidates were evaluated, but the historical targets are reused and no independent confirmation is claimed. At each refit, use the same 252-observation validation block and fit nuisance parameters strictly before it. Retain only validation origins whose complete 21-observation targets finish by the refit date. Minimize the equal-weight mean of normalized QLIKE at horizons 1, 5 and 21, using all eligible daily origins. A single selected specification serves all outer horizons.

Retain the earlier grids for d, η and τ, and add $b \in \{0, 0.5, 0.9\}$ and $\delta \in \{0, 1\}$. There are 288 candidates for sectors and 72 for the scalar index per kernel family. A symmetric variant fixes $\delta = 0$; a graph ablation fixes $\tau = 0$; a kernel ablation uses normalized exponential weights truncated to the same 252 observations, with the earlier four decay values. Each ablation has its own validation selection. The larger candidate set increases selection risk despite chronological ordering.

An additional GJR-GARCH reference permits asymmetric shock responses [12]. Its normalized variance satisfies $h_t = \omega + (\alpha + \gamma\mathbf{1}\{u_{t-1} < 0\})z_{t-1} + \beta h_{t-1}$. Independent asset fits use Gaussian QMLE, two fixed starts, nonnegative coefficients and $\alpha + \beta + \gamma/2 < 0.999$.

panel	index			sectors		
horizon	1	5	21	1	5	21
Original fractional	0.6514	0.4223	0.5467	0.6405	0.5627	0.6196
Persistent symmetric	0.6143	0.3815	0.5162	0.6447	0.5582	0.6116
Persistent asymmetric	0.5819	0.3574	0.4977	0.6242	0.5478	0.6043
Without graph	0.5819	0.3574	0.4977	0.6317	0.5474	0.6046
Exponential kernel	0.6223	0.3516	0.4785	0.6656	0.5588	0.5980
EWMA	0.8397	0.4576	0.5602	0.7195	0.5914	0.6251
GARCH	0.6004	0.3640	0.4714	0.6297	0.5499	0.6021
GJR-GARCH	0.5251	0.3128	0.4360	0.6053	0.5401	0.5931

Table 2: Mean normalized QLIKE for the refinement, its ablations and references. "Original fractional" denotes the initial variance specification. All rows use the same 186 origins. Lower is better.

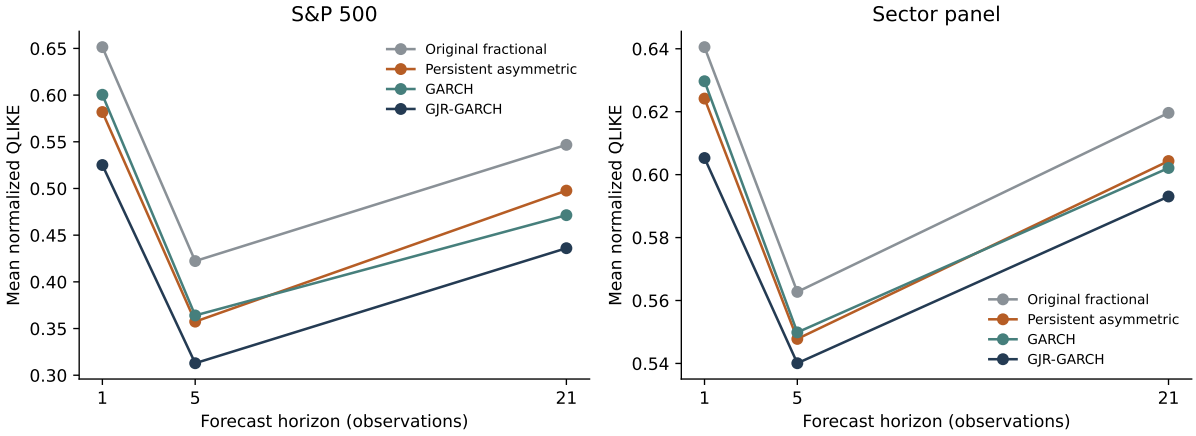


Figure 4: Pooled losses before and after the refinement, alongside GARCH and GJR-GARCH. The asymmetric reference remains stronger in every displayed case.

All 456 selected fits report convergence. Multi-step moments also require conditional symmetry. Earlier reference forecasts are recomputed; all 40,176 original forecast records and loss values match exactly, so comparisons use unchanged targets and scales.

7.4 Follow-up results

Table 2 shows lower pooled loss for persistent asymmetric fractional diffusion than the initial model in all six comparisons. Index scores change from $(0.65138, 0.42225, 0.54672)$ to $(0.58187, 0.35738, 0.49770)$, and sector scores from $(0.64053, 0.56274, 0.61963)$ to $(0.62417, 0.54779, 0.60432)$. The refinement also has lower loss than GARCH at one and five observations in both panels. GARCH remains better at 21 observations, while GJR-GARCH is better at every panel–horizon combination. These findings establish improvement within the studied fractional family without establishing dominance over asymmetric volatility references.

Using ten-origin blocks, nominal 95% ranges for the change relative to the initial fractional model exclude zero in five of six comparisons; the exception is one-session sector forecast-

ing. For the index, the one-session change is -0.06952 with range $[-0.14531, -0.00876]$. All ranges comparing the refinement with GARCH include zero. These descriptive ranges do not adjust for the repeated research on this snapshot and should not be read as independent confirmation.

Asymmetry is selected at all 38 index refits and 35 sector refits; nonzero variance persistence is selected at every refit. The graph is selected with nonzero strength at 18 sector refits. Relative to its graph ablation, the refinement improves the one-session sector mean by 0.00752, slightly worsens the five-session mean and changes the 21-session mean little. The exponential ablation is stronger for index horizons 5 and 21 and sector horizon 21. Thus neither graph smoothing nor the fractional kernel uniformly dominates its alternative. Annual improvements also vary: relative to the first specification, index loss rises at 21 sessions in 2024, and sector loss rises at 5/21 sessions in 2024 and at one session in 2025. All annual results remain in the public tables.

8 Extension to country ETFs and additional benchmarks

8.1 A specification fixed before the new download

The previous two studies reuse US history inspected during development. To assess transfer to another universe, a third protocol fixed eight country ETFs before downloading their prices: EWA (Australia), EWC (Canada), EWG (Germany), EWH (Hong Kong), EWJ (Japan), EWL (Switzerland), EWQ (France) and EWU (United Kingdom). These are long-established US-listed iShares equity funds, selected for geographical spread. The request covers 1 January 2018 through 16 October 2026, using the same adjusted-price convention. All eight assets are required; missing prices are not filled and assets are not removed after seeing results.

The unchanged fractional implementation, parameter grids, validation windows, refit schedule and three horizons are frozen in `paper/external_protocol.json`, with source hashes. No external loss is used to revise that specification. This is a locally timestamped protocol, not an independently registered preregistration. The country panel was not previously evaluated in this project, but it shares global shocks, US trading hours and dollar denomination with the development assets. It is therefore an extension of the asset universe on retrospective data. Shared shocks preclude treating it as an independent replication, and the experiment is not prospective.

A computational amendment is recorded in the accompanying protocol files. The first attempt reached the FIGARCH forecast and failed because a strided panel column was not C-contiguous. The correction copies input arrays to the required layout. It changes no model or evaluation rule; no FIGARCH comparison results existed at that point. The original protocol and failed-run record are retained.

8.2 Additional forecaster and fractional benchmark

The additional forecaster is deliberately fixed:

$$\widehat{V}^{\mathrm{ens}}_{o,h,j} = \tfrac{1}{2}\widehat{V}^{\mathrm{frac}}_{o,h,j} + \tfrac{1}{2}\widehat{V}^{\mathrm{GJR}}_{o,h,j}\,. \tag{23}$$

Here the fractional component is the persistent asymmetric model, including its selected graph diffusion. The coefficient is always one half, with no fitted or retrospectively selected combination weights. Positivity is preserved. The ensemble is an optional forecast combination, not a new generative fractional recurrence, and does not replace the standalone model. Its QLIKE is computed from the combined variance; component losses are not averaged.

Equal weighting gives each component the same contribution without estimating an additional parameter from this relatively short evaluation history. The weight was specified before the country-ETF download and is retained across assets and horizons. It is a transparent reference choice, not an estimated optimum. Learning a weight would require a further chronological validation layer and would introduce estimation uncertainty. Since GJR is already a strong component, improving on the fractional model alone is insufficient evidence of complementarity: the relevant comparison also includes GJR by itself. Both components are therefore reported alongside their combination.

FIGARCH(1,d,1) [2] is added as an estimated benchmark using `arch` 7.2.0, Gaussian QMLE and the library's 1,000-lag ARCH approximation. Each asset is centered and standardized using the same origin-fitted mean and scale as the other methods. Fit on the preceding 1,008 returns at the same refit dates, with default starting estimates and SLSQP limited to 1,000 iterations. Between refits, keep parameters and the fitting backcast fixed while updating on observed returns. Analytic forecasts are transformed back to asset units and summed over the same horizons. Nonfinite results abort the experiment; finite nonconverged fits are reported rather than silently replaced. FIGARCH's approximation and initialization differ from our normalized 252-lag kernel and are recorded explicitly.

The same ensemble and FIGARCH reference are appended to the recorded US forecasts after checking their code and input hashes. Those comparisons remain exploratory. Every table uses identical targets within its panel; pooled, annual and asset-level losses are retained. Paired uncertainty ranges use the same whole-panel circular bootstrap and the same nominal interpretation as the earlier study.

8.3 Results

The international panel contains 2,210 prices per asset. Evaluation has 186 origins from 6 January 2023 to 16 September 2026, with complete targets through 15 October 2026 and 38 parameter refits. Its five-observation grid has a different calendar phase from the US grid; comparisons are paired within panels. All 760 FIGARCH asset fits across the three panels converge, as do all 304 international GARCH and 304 GJR-GARCH fits.

The unchanged fractional model obtains international QLIKE $(0.52898, 0.66359, 0.67147)$ at horizons $(1, 5, 21)$, below both EWMA and FIGARCH at every horizon. However, GARCH and GJR-GARCH are stronger at all three. The no-graph ablation has lower

horizon	1	5	21
EWMA	0.5321	0.6700	0.6823
GARCH	0.5287	0.6587	0.6672
GJR-GARCH	0.5083	0.6496	0.6623
FIGARCH	0.5438	0.6671	0.6730
Fractional	0.5290	0.6636	0.6715
Without graph	0.5234	0.6633	0.6701
Exponential kernel	0.5290	0.6526	0.6653
Fixed 50:50 ensemble	0.5131	0.6509	0.6613

Table 3: Mean normalized QLIKE on the eight pre-specified country ETFs. The fractional row is the unchanged persistent asymmetric model. The fixed ensemble averages its variance forecasts with GJR-GARCH. Lower is better.

loss throughout, while the exponential kernel is better at horizons 5 and 21. Thus the second universe does not support a general benefit from diffusion or a uniform fractional-kernel advantage.

The fixed ensemble lowers those fractional scores to $(0.51307, 0.65094, 0.66134)$, improving all three external pooled comparisons. Its asset-level loss is lower in 22 of 24 country–horizon comparisons, but annual pooled loss worsens relative to the standalone fractional model at horizons 5 and 21 in 2024. Across the US and international panels, the ensemble improves on both the standalone fractional model and FIGARCH in all nine panel–horizon comparisons. These are loss differences, not percentages of an arbitrarily normalized score.

Relative to GJR-GARCH, the ensemble wins only for US sectors at 5 and 21 observations and for the international panel at 21 observations. The international 21-session difference is -0.00092, with ten-origin-block nominal 95% range $[-0.01174, 0.01110]$. All nine ensemble-versus-GJR ranges include zero. For international one-session forecasts, the change from the standalone fractional model is -0.01590; the interval is $[-0.03100, -0.00219]$.

Compared with FIGARCH, the difference is -0.03073. Its interval runs from -0.05536 to -0.01145. These descriptive intervals support reporting the observed improvement without claiming reliable superiority over GJR-GARCH. Full annual, asset-level and paired comparisons remain in the public outputs.

8.4 Direct comparison with Network GARCH

A follow-up benchmark evaluates the first-order recursion of Zhou et al. [6], equation (2.2), on the same sector and country panels:

$$h_t^N = \omega \mathbf{1} + \alpha z_{t-1} + \lambda B z_{t-1} + \beta h_{t-1}^N, \qquad \omega > 0, \quad \alpha, \lambda, \beta \geq 0, \quad \alpha + \lambda + \beta < 0.999. \tag{24}$$

Here B row-normalizes the positive squared correlation weights already used to construct the fractional model's graph. Its diagonal and isolated rows are zero. A sensitivity benchmark replaces these weights by binary edges at $R_{ij} \geq 0.5$ before normalization; a pooled GARCH control sets $\lambda = 0$. The coefficients are shared across the standardized assets. These choices adapt the published recursion to estimated correlation networks

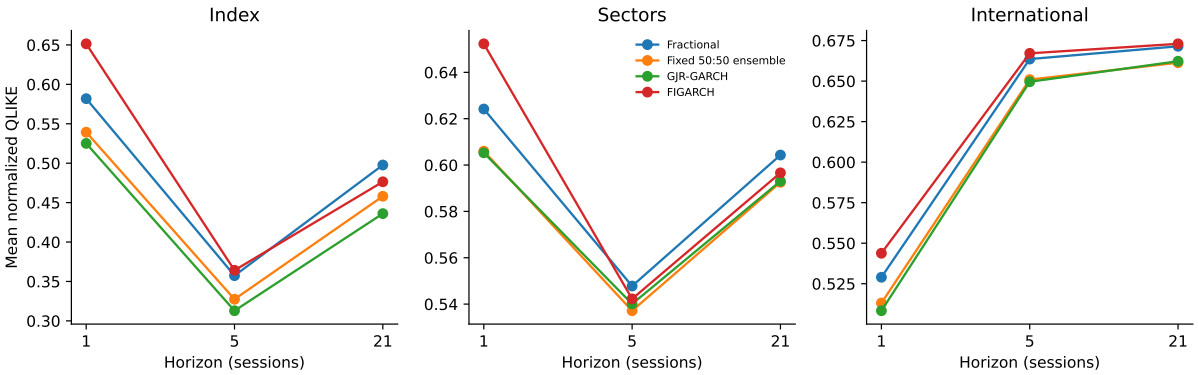


Figure 5: Standalone fractional forecasts, the fixed ensemble, GJR-GARCH and FIGARCH. The index and sector panels reuse the development snapshot; the international panel follows the separately recorded external-universe protocol.

rather than the original shareholder network. They do not reproduce the original paper's empirical application or its assumptions of an exogenous network.

At each of the 38 retained refits per panel, marginal Gaussian QMLE uses the preceding 1,008 returns, the common fitted means and scales, and initial normalized variance one. Three fixed starting points are optimized with analytic gradients; the fit with the lowest finite training objective is retained. Observed returns update the state between refits. The first future variance follows (24); subsequent conditional expectations replace future z_t by its conditional mean h_t^N. Daily forecasts are summed over the original horizons and evaluated on identical targets. All 228 selected fits and all 684 optimization starts converge.

The new protocol was recorded after examining the earlier results and before fitting these benchmarks. It is an exploratory comparison on reused data; the fractional forecasts, combination weights and previous studies are unchanged. The index is excluded because a single asset supplies no network comparison. Annual scores, asset-level scores and all fitting attempts are retained with the protocol.

	sectors			international		
	1	5	21	1	5	21
Fractional	0.6242	0.5478	0.6043	0.5290	0.6636	0.6715
Fixed 50:50 ensemble	0.6060	0.5371	0.5925	0.5131	0.6509	0.6613
GJR-GARCH	0.6053	0.5401	0.5931	0.5083	0.6496	0.6623
FIGARCH	0.6523	0.5422	0.5966	0.5438	0.6671	0.6730
Network GARCH (weighted)	0.6149	0.5465	0.6004	0.5397	0.6642	0.6741
Network GARCH (binary)	0.6172	0.5490	0.6029	0.5401	0.6648	0.6750
Pooled GARCH	0.6321	0.5513	0.5997	0.5296	0.6614	0.6725

Table 4: Mean normalized QLIKE for the separately recorded network comparison. The columns retain the same 1-, 5- and 21-session targets. Network GARCH uses shared coefficients; the pooled control removes its network term. Lower is better.

The weighted Network GARCH benchmark has lower pooled loss than the standalone fractional model at all three sector horizons; the ordering reverses at all three country horizons. Every paired 95% range for these six differences includes zero at both block lengths. The fixed ensemble has lower pooled loss than either network variant and the pooled control in all six comparisons. Its one-session country difference against weighted Network GARCH is −0.02660, with ten-origin-block range [−0.04645, −0.00747]; the five-session range excludes zero with ten-origin blocks but includes it with five-origin blocks. The other four ranges include zero at both lengths.

Relative to pooled GARCH, weighted network feedback helps the sector mean at one and five sessions, but worsens it at 21 sessions and at all country horizons. All six paired ranges include zero. Binary network losses are higher than weighted network losses in each comparison. This additional experiment therefore supports a context-dependent role for cross-asset feedback while leaving GJR-GARCH a strong reference.

8.5 Observed evolution and rolling predictions

Aggregate losses summarize accuracy but do not show how forecasts respond over time. Figures 6 and 7 compare the recorded predictions with the subsequently observed squared-return targets over the full evaluation period. For readability, both are expressed in annualized volatility units:

$$\widehat{\sigma}^{\mathrm{ann}}_{o,h,j} = 100\sqrt{\frac{252}{h}\widehat{V}_{o,h,j}}, \qquad \sigma^{\mathrm{proxy}}_{o,h,j} = 100\sqrt{\frac{252}{h}Y_{o,h,j}}. \tag{25}$$

These are square-root transformations of variance forecasts and noisy targets, not estimates of an observed latent volatility path. In particular, the square root of expected variance is not generally expected volatility. At one session, the observed series is a scaled absolute centered return and is especially noisy.

The new time-series panels use the *end of the target window* on the horizontal axis. A prediction plotted at $o+h$ was issued at o, using only the data available then. Curves join the retained origins, spaced five observed sessions apart; they are not a new forecast issued every day. No further fitting, time smoothing, clipping of peaks or favorable subperiod selection is applied. The 21-session target windows overlap.

For the multi-asset displays, average $\widehat{V}_{o,h,j}$ and $Y_{o,h,j}$ over all assets before applying (25). Averaging volatilities directly would give a different quantity. The panel curves can also mask opposing asset-level errors. Appendix D therefore retains the complete set of 21-session paths for every sector and country ETF. These visual diagnostics complement the common-target losses and paired comparisons; visual tracking alone does not establish predictive superiority.

The index curves also reveal limitations hidden by pooled scores. The abrupt 2025 increase is incompletely anticipated, and the persistent forecasts can remain elevated after the observed proxy falls. Smoother predictions should not be interpreted as pointwise reproduction of the realized fluctuations.

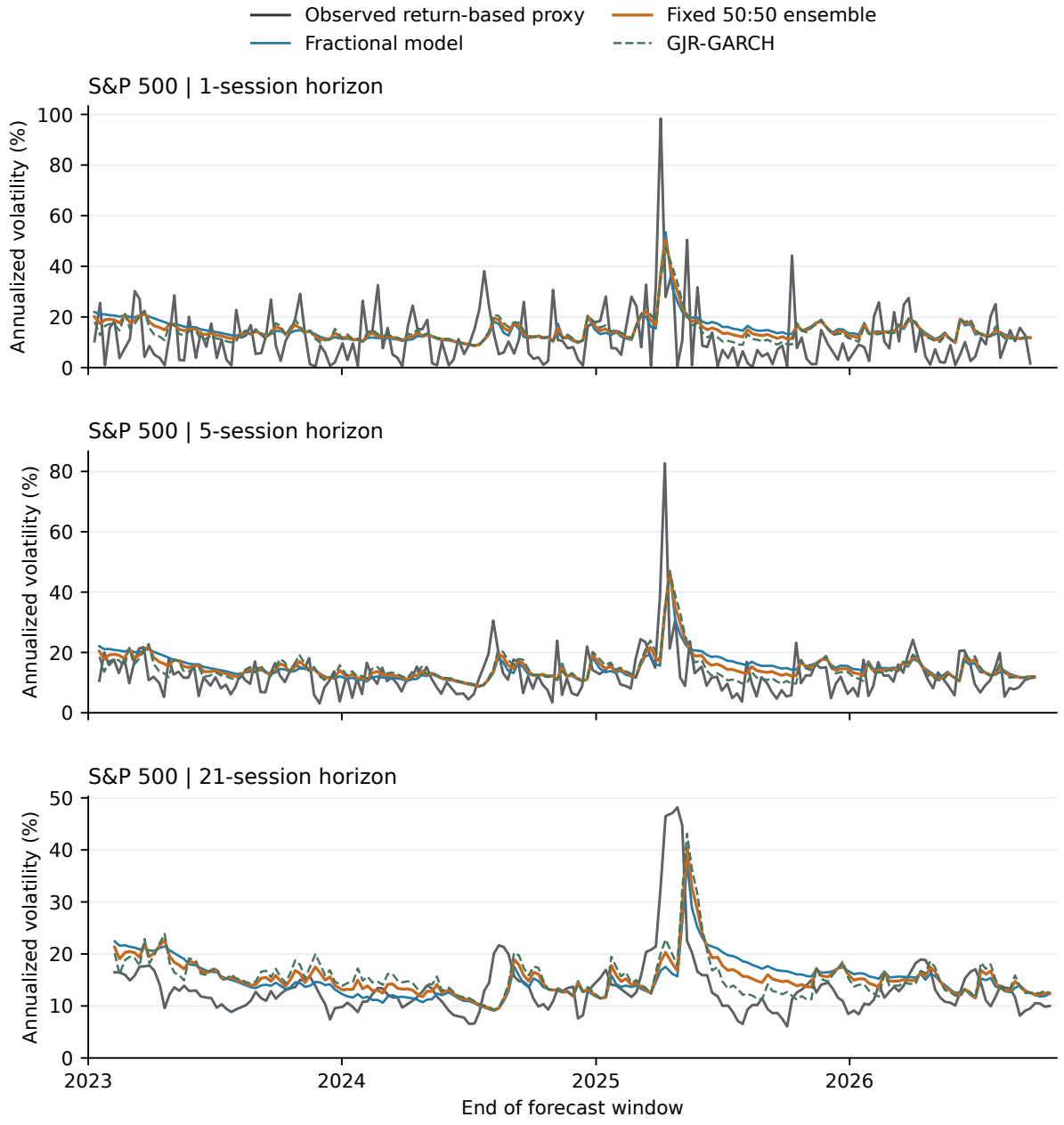


Figure 6: S&P 500 observed proxy and rolling predictions at 1, 5 and 21 sessions. Black denotes the subsequent return-based observation; blue is the persistent fractional model, orange the fixed ensemble, and dashed green GJR-GARCH. Each forecast was issued the stated number of observations before its plotted window-end date. Vertical scales differ by horizon to retain the full peaks.

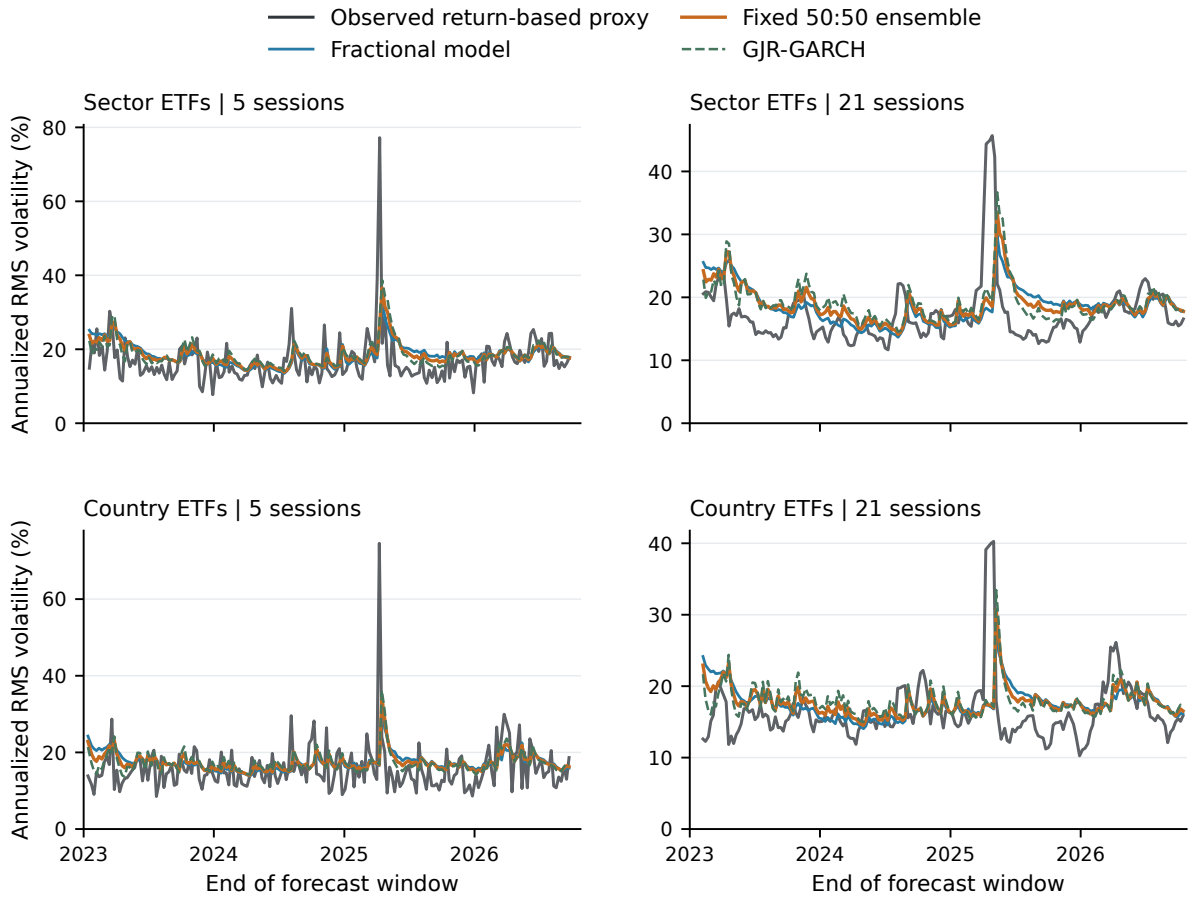


Figure 7: Observed and predicted risk evolution for all eleven sector ETFs and all eight country ETFs, at horizons of 5 and 21 sessions. Marginal variances are averaged across the complete panel before taking the square root. These are annualized root-mean-square marginal volatilities, not portfolio volatility; cross-asset covariances are not included in this display.

9 Interpretation, limitations and conclusions

Fractional weighting and graph diffusion provide an interpretable structure for a positive conditional-variance model. Its shared temporal profile and spatial resolvent keep the parameterization compact, while the fixed-parameter moment arguments and exact forecast recursion make its behavior explicit. This is a discrete variance specification; the original price simulator remains separate and is not justified by a fractional pricing PDE.

The clearest empirical improvement comes from persistence and asymmetric responses, which lower loss relative to the initial fractional specification in all six US comparisons. Combining the refined forecasts with GJR lowers pooled QLIKE further and outperforms FIGARCH in all nine panel–horizon comparisons. The direct network benchmark adds a useful contrast: weighted Network GARCH performs better in the sector panel, whereas the fractional model performs better in the country panel. Uncertainty ranges do not resolve these differences. GJR remains difficult to improve upon, and the country extension provides no general evidence that network feedback helps.

Interpretation depends on the evaluation design. The panels share market shocks and use retrospective adjusted prices; moving to country ETFs extends the universe without providing independent or prospective confirmation. Squared daily returns also leave considerable noise in the risk target. The fixed memory window, shared graph operator and constant conditional mean restrict the dynamics that can be represented. These constraints help explain the model and delimit the questions answered by the experiments.

A next study should prioritize point-in-time inputs, subsequently observed targets and richer volatility measurements. Comparing estimated network log-ARCH, threshold Network GARCH and dynamic-covariance models would further clarify which restrictions matter. The present evidence supports the fractional–diffusive specification as a transparent component for risk forecasting research, with mixed evidence for its network contribution.

A Implementation and reproduction

The initial implementation is in `fractional_diffusion.py`; the benchmark and asset generator are in `run_paper.py`. The follow-up uses `refined_fractional.py` and `run_refinement.py`. The country-ETF extension uses `external_validation.py`, `download_external.py` and `run_external.py`.

The network comparison is implemented in `network_benchmark.py`. Run it with `run_network_benchmark.py`; its separate protocol is `paper/network_protocol.json`.

Existing return simulators and forecast heads remain separate. The recorded protocol is `paper/protocol.json`. Input hashes, candidate scores, graph estimates, selected parameters, optimizer statuses and each origin's forecasts are saved. Raw prices and detailed forecast records remain in ignored local artifacts.

```
python run_paper.py --synthetic --output artifacts/paper-synthetic
python run_paper.py --data-dir artifacts/yahoo-through-2026-10-16 \
  --output artifacts/paper-study --publish-assets
python run_refinement.py --synthetic --output artifacts/refinement-synthetic
```

```
python run_refinement.py --data-dir artifacts/yahoo-through-2026-10-16 \
  --output artifacts/refinement-study --publish-assets
python -m unittest discover -s tests -v
cd paper
tectonic --reruns 3 main.tex
```

The synthetic command uses 1,300 correlated Gaussian price observations with no imposed time-varying volatility or fractional mechanism. It is an offline execution example, not empirical evidence for the proposed model. The market snapshot download command and adjustment convention are documented in the data guide. External-study commands and source-package instructions are in `paper/README.md`. The build uses ordinary LaTeX article typography, numbered equations, a contents page and running headers, preserving the presentation of the original project report.

The combined suite contains 61 tests on Python 3.12 under macOS, including Network GARCH gradient checks, conditional-moment simulation comparisons, asset relabeling and future-price perturbations. Checks cover fractional weights, graph positivity and contraction, asset relabeling, GARCH/GJR filtering, exact moments against Monte Carlo, asymmetric normalization and validation forecasts against individually truncated histories. Future-price perturbations verify that changing unobserved observations cannot alter earlier variances, selected parameters or graphs. Additional checks cover FIGARCH library equivalence on strided panel inputs, matched ensemble scoring and the effective-memory impulse response. Numerical checks support implementation fidelity; mathematical claims rest on their stated assumptions and proofs.

The presentation script `render_prediction_figures.py` regenerates the direct-score comparison and observed-versus-forecast panels from saved records, without refitting models. It is run after the numerical exporters and before compiling this manuscript; input and output hashes accompany the figures.

B Annual scores

panel	index			sectors		
horizon	1	5	21	1	5	21
year						
2023	0.0021	-0.0541	-0.0447	0.2066	0.1643	0.1675
2024	0.8771	0.2769	0.3731	0.3427	0.3209	0.3531
2025	1.2053	0.9798	1.3312	1.1208	0.9419	1.1219
2026	0.4703	0.5115	0.5196	0.9897	0.9254	0.9201

Table 5: Annual mean normalized QLIKE for the fractional graph model. Levels reflect both realized fluctuations and the origin's reference scale. Negative scores are permissible; comparisons should use common-origin loss differences.

C Relation to the original model

The original project used signed Grünwald–Letnikov coefficients in a return recurrence and a reflected nearest-neighbor sector Laplacian. Isolating the current value in a GL derivative does not yield that recurrence's lag and sign convention. A fractional derivative also does not inherit the ordinary logarithmic chain rule. The present work therefore starts from (4)–(10), rather than invoking a fractional Black–Scholes equation.

The retained idea is memory plus diffusion. The changes are explicit: positive fractional-integral weights act on squared standardized innovations; a symmetric estimated graph replaces the artificial sector lattice; positive feedback controls second moments; and risk forecasts are compared chronologically against strong variance references. Hurst identification, convergence of a pricing PDE, and directional price-prediction superiority are not claimed.

D Observed and predicted paths for every ETF

The following figures show every ETF in the two recorded panels, using the 21-session horizon and the full evaluation period. Splitting the sector list across two pages only improves readability; assets are neither selected nor ranked by forecasting performance. The horizontal coordinate is the target window end, and the vertical coordinate is given by (25). Each asset has its own vertical scale, starting at zero. No peak is clipped, and no extra temporal smoothing is applied.

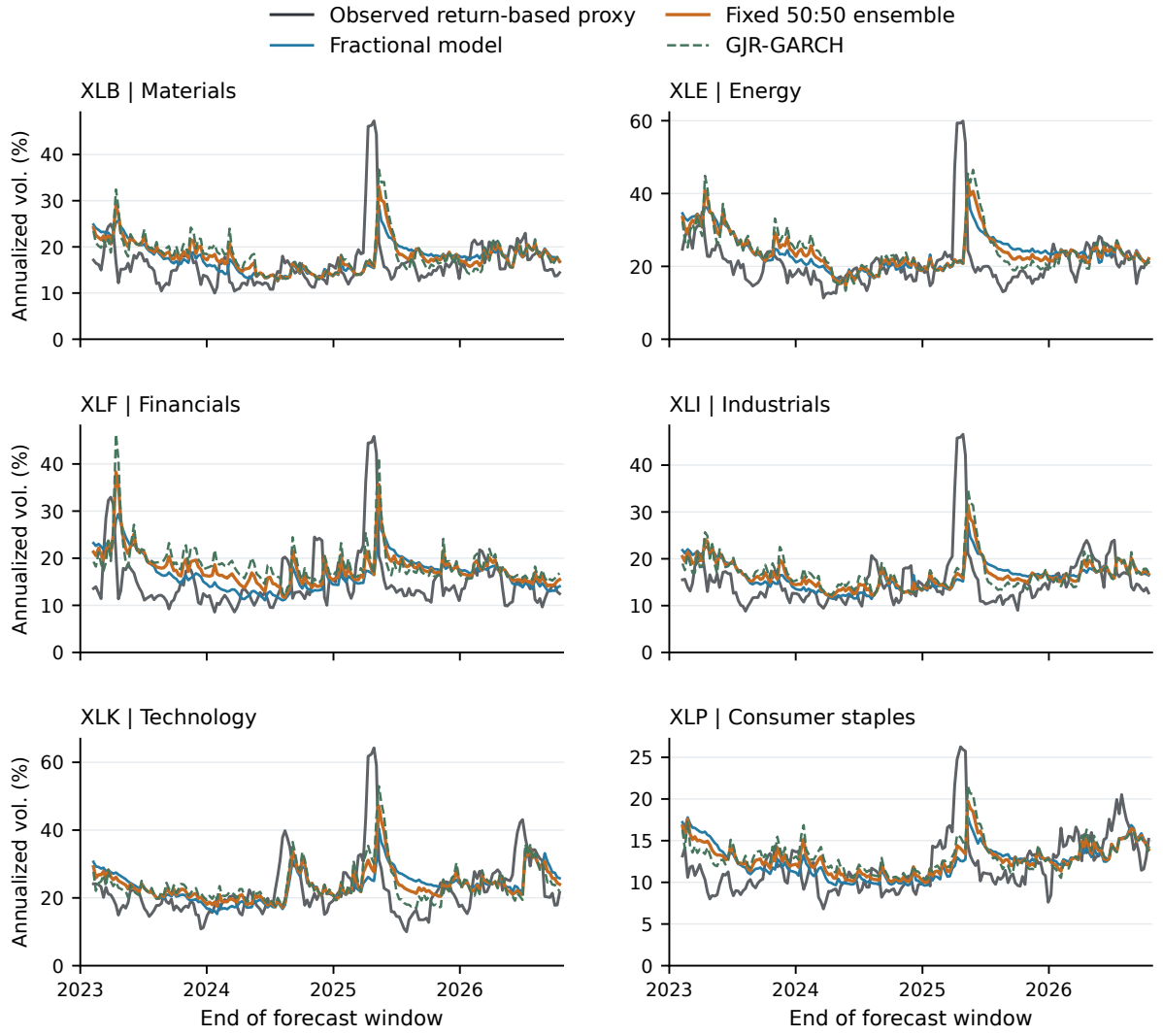


Figure 8: Twenty-one-session observed proxies and forecasts for materials, energy, financials, industrials, technology and consumer staples. Curves use the same methods and colors as Figure 6.

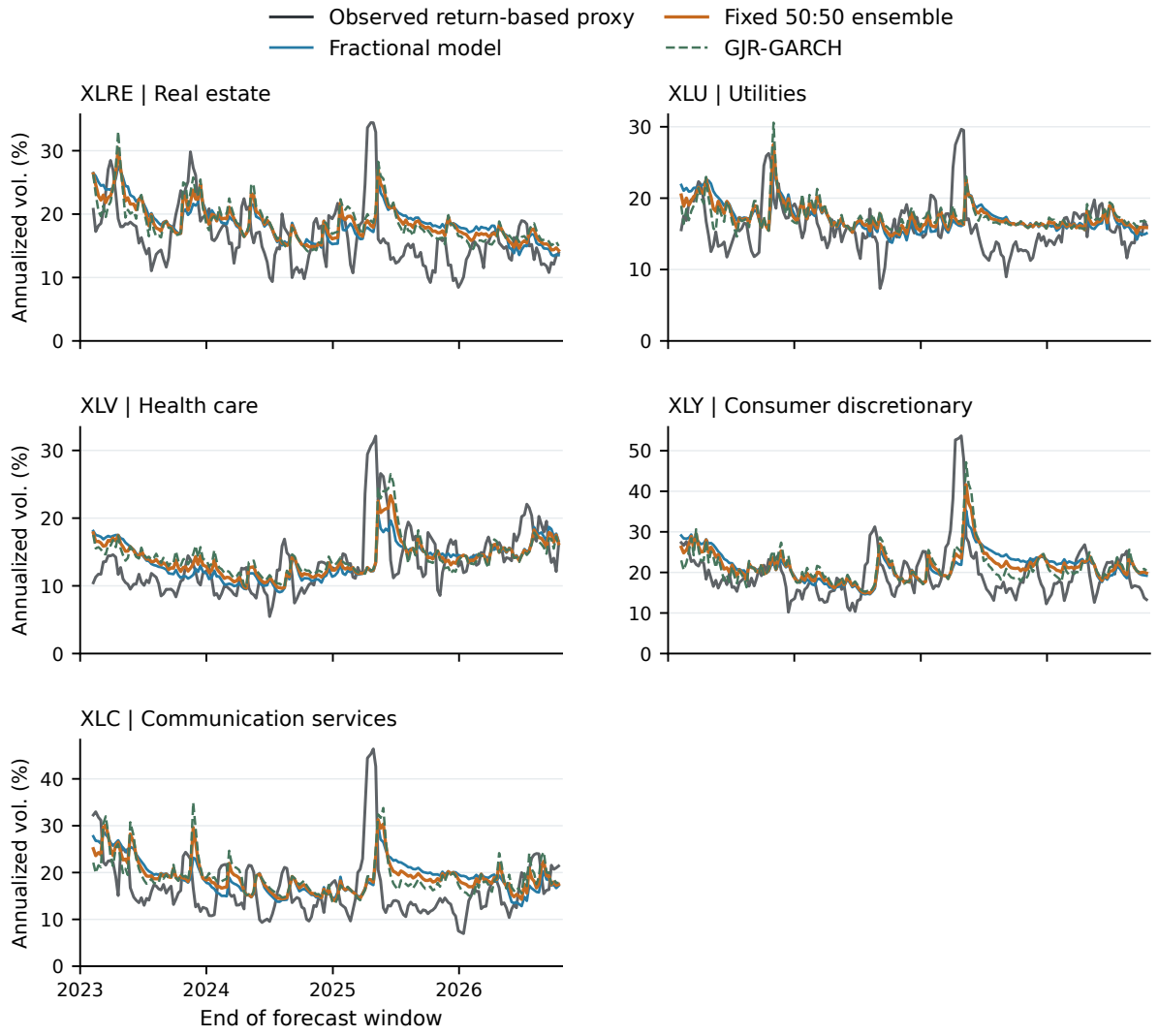


Figure 9: The remaining five sector ETFs: real estate, utilities, health care, consumer discretionary and communication services. Every sector is included exactly once across these two figures.

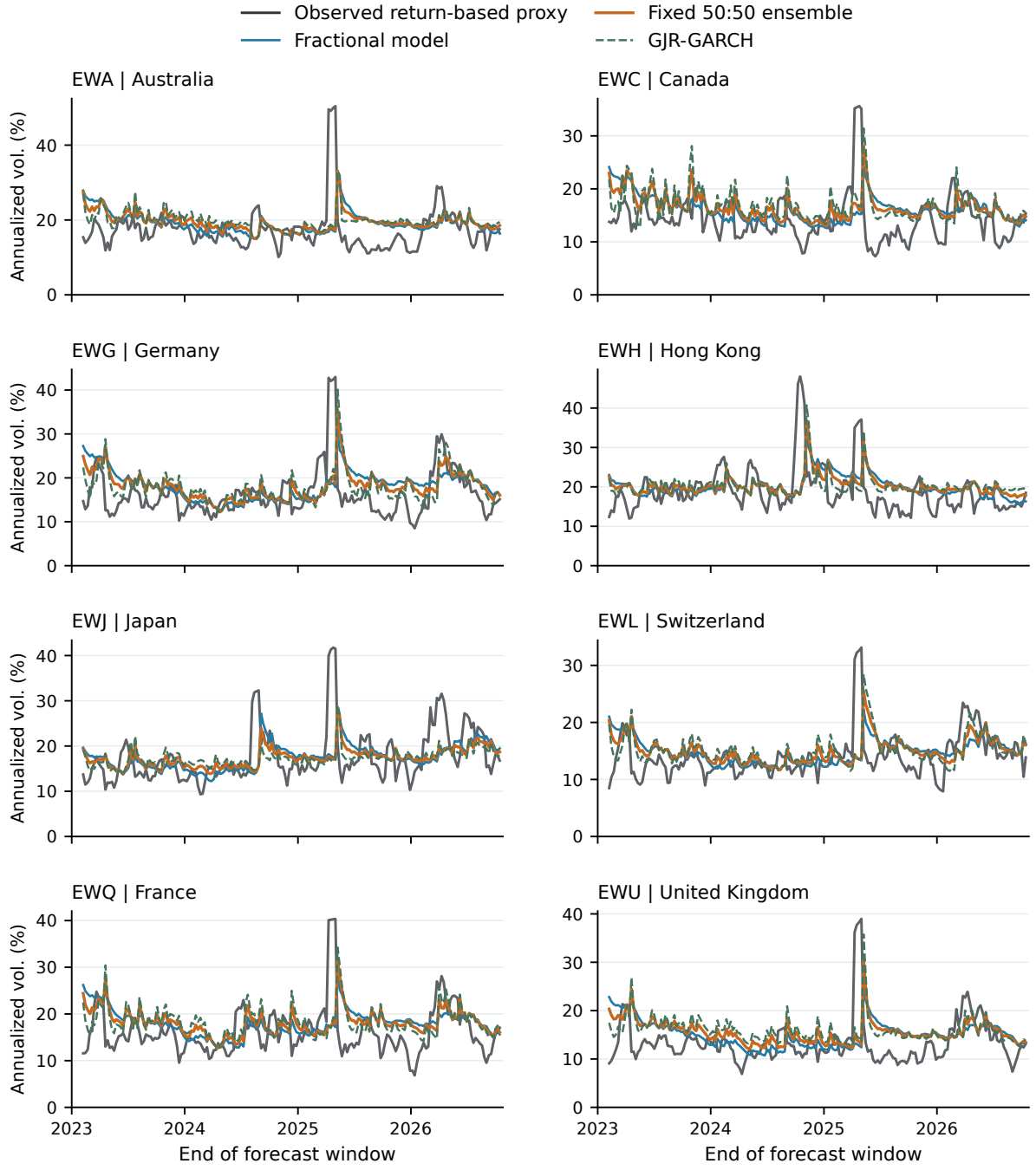


Figure 10: Twenty-one-session observed proxies and forecasts for all eight pre-specified country ETFs. The observed series is a noisy daily-return-based proxy; the plots do not imply prediction of the underlying ETF price direction.

References

[1] J. R. M. Hosking. Fractional differencing. *Biometrika*, 68(1):165–176, 1981. doi: 10.1093/biomet/68.1.165.

[2] Richard T. Baillie, Tim Bollerslev, and Hans Ole Mikkelsen. Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1):3–30, 1996. doi: 10.1016/S0304-4076(95)01749-6.

[3] David I. Shuman, Sunil K. Narang, Pascal Frossard, Antonio Ortega, and Pierre Vandergheynst. The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains. *IEEE Signal Processing Magazine*, 30(3):83–98, 2013. doi: 10.1109/MSP.2012.2235192.

[4] Philipp Otto, Wolfgang Schmid, and Robert Garthoff. Generalised spatial and spatiotemporal autoregressive conditional heteroscedasticity. *Spatial Statistics*, 26:125–145, 2018. doi: 10.1016/j.spasta.2018.07.005.

[5] Raffaele Mattera and Philipp Otto. Network log-ARCH models for forecasting stock market volatility. *International Journal of Forecasting*, 40(4):1539–1555, 2024. doi: 10.1016/j.ijforecast.2024.01.002.

[6] Jing Zhou, Dong Li, Rui Pan, and Hansheng Wang. Network GARCH model. *Statistica Sinica*, 30(4):1723–1740, 2020. doi: 10.5705/ss.202018.0234. URL https://www3.stat.sinica.edu.tw/statistica/j30n4/j30n403/j30n403.html.

[7] Yue Pan and Jiazhu Pan. Threshold network GARCH model. *Journal of Time Series Analysis*, 45(6):910–930, 2024. doi: 10.1111/jtsa.12743.

[8] Peiyi Zhou. A new implementation of network GARCH model for stock volatility and co-volatility forecasting. *Journal of Forecasting*, 2026. doi: 10.1002/for.70191. Published online 12 July 2026.

[9] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.

[10] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.

[11] Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.

[12] Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5):1779–1801, 1993. doi: 10.1111/j.1540-6261.1993.tb05128.x.